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Lab Manual : **7(String)**
Course Code : **CSE103**
Course Title : **Structured Programming**
Instructor : Dr. Sajeeb Saha, Adjunct Faculty, Department of CSE

Objective:

The objective of this lab is to provide a fundamental idea about string operations of C programming. At the end of the lab, students are able to know:

- How to declare string variable and assign values to variable or take input
- How to find the length of string
- How to solve different problem such as concatenation, copy, compare using string
- How to handle string using ASCII code

Preliminary:

Strings are actually one-dimensional array of characters terminated by a **null** character '\0'. Thus a null-terminated string contains the characters that comprise the string followed by a **null**.

The following declaration and initialization create a string consisting of the word "Hello". To hold the null character at the end of the array, the size of the character array containing the string is one more than the number of characters in the word "Hello."

```
char greeting[6] = {'H', 'e', 'l', 'l', 'o', '\0'};
```

Exercise 1

Strlen() returns the length of the string without including end character **(terminating char '\0')**.

```
#include<stdio.h>
#include<string.h>
int main(){
    int len;
    char str[100];
    gets(str);
    len = strlen(str);
    printf("Length of string = %d\n",len);
    return 0;
}
```

Write a C program to find length of a string using loop.

```
#include <stdio.h>
#include <string.h>

int main()
{
  char Str[100];
  int Length = 0;

  printf("\n Please Enter any String \n");
  gets(Str);

  while (Str[Length] != '\0')
   {
     Length++;
   }

  printf("\nLength of a String = %d", Length);
  return 0;
}
```

Example:
Input string: I love programming. I love CodeforWin.
Output length of string: 38

Exercise 2

The strcpy() function copies one string to another string.

```
#include<stdio.h>
#include<string.h>
int main(){
    int len;
    char str1[100], str2[100];
    gets(str1);
    strcpy(str2,str1);

    printf("Input String1 = %s\n",str1);
    printf("Output String2 = %s\n",str2);

    return 0;
}
```

Example:
Input string1: Hello World!
Output string2: Hello World!

Write a C program to copy one string to another string without using built-in function.

```
#include<stdio.h>

int main() {
   char s1[100], s2[100];
   int i;

   printf("\nEnter the string :");
   gets(s1);

   i = 0;
   while (s1[i] != '\0') {
      s2[i] = s1[i];
      i++;
   }

   s2[i] = '\0';
   printf("\nCopied String is %s ", s2);

   return (0);
}
```

Exercise 3

Write a C program to concatenate two different strings into single string without using built-in function.

```
#include <stdio.h>
#include <string.h>

int main()
{
        char Str1[100], Str2[100];
        int i, j;

        printf("\n Please Enter the First String :  ");
        gets(Str1);

        printf("\n Please Enter the Second String :  ");
        gets(Str2);

        i = 0;
        while( Str1[i]!='\0')
        {
                i++;
        }

        j = 0;
        while( Str2[j]!='\0')
        {
                Str1[i] = Str2[j];
                i++;
                j++;
        }
        Str1[i] = '\0';

        printf("\n String after the Concatenate = %s", Str1);

        return 0;
}
```

Example:
Input string1: I love
Input string2: CodeforWin
Output string: I love CodeforWin

Exercise 4

Write a C program to compare two strings using loop character by character without using built-in function

Example:
Input string1: CodeforWin
Input string2: CodeforWin
Output: Both strings are equal

Exercise 5

Write a C program to convert string from lowercase to uppercase string using loop
Example:
Input string: I love programming!
Output string: I LOVE PROGRAMMING!

Exercise 6

Write a C program to count total number of words in a string using loop

Example:
Input string: I love C programming!
Total number of words: 4

Exercise 7

Write a C program to count total number of count vowels, consonants, number and others character etc.

Suppose you have a char array namely line[]. Then you can write a if statement to count vowel using the following condition:

```
// check if the character is a vowel
    if (line[i] == 'a' || line[i] == 'e' || line[i] == 'i' ||
        line[i] == 'o' || line[i] == 'u'){

        vowel++;
}
// check if the character is a digit
    else if (line[i] >= '0' && line[i] <= '9') {
      digit++;
    }
```

Exercise 8

Write a program in C to separate individual characters from a string.
Test Data :
Input the string : eastwest

Expected Output :
The characters of the string are :
e a s t w e s t

Exercise 9

Write a program in C to print individual characters of a string in reverse order.

Test Data :
Input the string : eastwest

Expected Output :
The characters of the string in reverse are :
tsewtsae

Exercise 10

Write a program in C to count the total number of alphabets in a string.

Test Data :
Input the string : Welcome to ewu

Expected Output :
Number of Alphabets in the string is : 12

Exercise 11

Write a program in C to find the maximum number of characters in a string.

Test Data :
Input the string : Welcome to ewu

Expected Output :

The Highest frequency of character 'e'
appears number of times : 3

Exercise 12

Write a program in C to check whether a string is palindrome or not.

Test Data :
Input the string : madam i m adam

Expected Output :
The string is a palindrome

Test Data :
Input the string : Hello

Expected Output :
The string is not a palindrome